

One type of worker I study is **platform-based gig workers**, whose labor and services are increasingly woven into the fabric of everyday life. For consumers, gigs offer convenient, on-demand help with everyday tasks – e.g., Uber, caretaking, creative and professional projects – by drawing on the wide-ranging expertise of a global workforce. For workers, the temporary and task-based nature of gigs represents *independence* from traditional jobs and organizations – affording them the perceived freedom to choose not only *when* and *where* to labor, but also *what* to work on. However, as platforms mediate between workers and consumers, their hegemonic data practices shift significant agency and autonomy away from both stakeholder groups [a]. To centralize power, platforms capitalize on the absence of legal protections for independent contractors to exploit gig workers physically, financially and psychologically [b]. With rising rates of worker turnover and displacement, how do we create transformative interventions that propel workforces toward more natural, collective and liberated future interactions? Overall, the primary themes of my research:

- ① employs large-scale computational methods to identify worker practices [1] and intentions [11]
- ② co-designs with multi-stakeholder communities to envision collective, sustainable futures [2, 5, 6]
- ③ develops [7] interventions that guide more joyful [10], pluralistic [3, 9] and natural interactions.

① Understanding Worker Practices & Rationales via Large-Scale Data Analysis [1, 11]

Gig platforms commonly leverage a consumer-first rhetoric to justify and avoid accountability for unfair labor practices. But despite highly competitive, isolating platform environments, workers continue to willingly engage – suggesting the presence of underexamined intrinsic motivations that drive them to develop strategies for handling unexpected challenges. In my first pair of contributions, I present large-scale analyses that uncover (1) *how* freelancers communicate on platforms to achieve productivity and success (Fig. 1) as well as (2) crowdworker rationales on *why* they perform platform-mediated gigs despite harsh working conditions.

To **scalably understand individual worker strategies**, we engaged in multi-month negotiations with collaborators to access a large-scale dataset (> 2 million messages, >58k freelancers, >56k projects) from a leading online freelancing platform. Applying regression analyses on these platform-mediated freelancer-client communications, I found statistically significant correlations between (1) personalized message content and higher *job acquisition rates* (2) standardized messaging schedules and increased *job completion rates*, as well as (3) repeated template use across projects that associated with greater bid volumes and *long term revenue* [1].

But *why* are workers motivated to continue engaging in online platform labor despite subpar conditions? Most workers' underlying beliefs about platforms likely remained undisclosed, under-examined, and misguided by platform-constructed illusions of flexibility. To explore such latent *grievances* and *rationales*, we leveraged the persuasive power of language models to intervene and dissuade crowdworkers from believing five common myths around gig work conditions. In our sample, 201 of 263 Prolific crowdworkers believed at least one myth, and by eliciting their rationales and responding with persuasive LLM responses, we revealed how **crowdworkers labor out of intrinsic desires to preserve independence, autonomy and flexibility** [11].

② Co-designing Multi-Stakeholder Interventions for Platform Accountability [2, 5, 6, 7]

Mobilizing the data and efforts of online gig workforces revealed crucial strategies and rationales. But offline gig work requires support from higher-power stakeholders to enact infrastructural reforms that advance gig labor rights and protections. To synergistically integrate priorities from multiple stakeholders, I led co-design workshops with policy experts, platform employees and workers to tackle five representative problems in gig work [2]. Our solution-oriented version of the speed-dating design method led stakeholders to generate and rank infrastructural interventions they are most prepared or motivated to implement (Fig. 2), which include visions of *stronger regulation*, *worker-owned coops* and *platform-government co-regulation*. Stakeholders also envisioned more incremental advances: platform-driven services/programs and (collectivist) technologies to enhance scheduling, surface resources and support financial well-being.

Uncovering collective visions for infrastructural and regulatory change is a critical first step, but to achieve these in practice? In a CSCW ‘24 workshop, I led like-minded scholars to explore *worker data collectives* [6] for enacting counterdata production. I also interviewed policy experts from varying levels of office to surface prioritized initiatives for data-sharing and design recommendations [5] that subsequently informed early-stage designs of [Gig2Gether](#) (Fig. 3), an open-sourced, cross-platform (pink-, white- and blue-collar domains) web app allowing users to track and share gig work data, view aggregate data visualizations and post qualitative stories about labor conditions that promote evidence-based advocacy and policymaking [6]. In a 7-day field evaluation with 16 participants, **Gig2Gether facilitated cross-platform mutual support** and financial reflection/planning, in addition to revealing latent worker desires for future use cases — including *collective algorithmic auditing and policy advocacy*. For one user, Gig2Gether represented “*a door between [gig workers] and politicians*”.

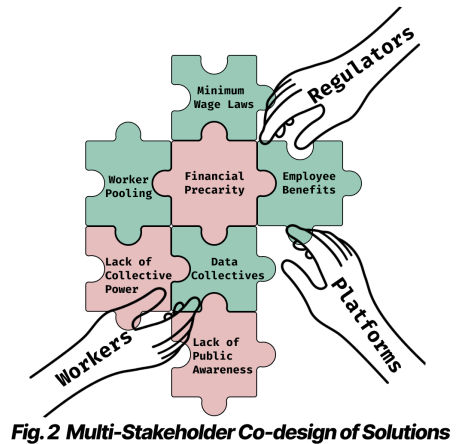


Fig. 2 Multi-Stakeholder Co-design of Solutions

③ Building Social Solidarity & Pluralistic Interactions amidst Evolving Tech [3, 10]

Alongside technological interventions [2], regulators in our workshops also called for increased and individualized policy. In response, I synthesized legal and policy documents, labor regulations, exemplary state bills and HCI scholarship to envision advances. In an award-winning position paper [3], I advocated for targeted and pluralistic policies that consider diverse task domains and intersectional worker backgrounds. But while policymakers call for stronger labor regulations, platforms continuously rely on workers’ non-employee status to evade employer accountabilities – causing them to shoulder *vulnerabilities like immaterial labor and emotional burdens, alone* [c]. Meanwhile, HCI research has overlooked the role of consumers in worker advocacy, which harms workers when uninformed consumers hold unrealistic expectations of service quality.

How to design socially approachable interventions to **motivate consumer-worker solidarity**? To catalyze *worker-consumer co-liberation* for ridehail, I explored how gamification can motivate passenger understanding and behavioral change around latent driving conditions [10]. Probing 15 passengers and 19 drivers with six interactive in-ride games during workshops, I uncovered passenger knowledge gaps (*e.g.* long-term health impacts, rating-imposed service expectations) as well as desires for more socially appropriate passenger-driver interactions from both stakeholders, opening opportunities for consumer-led advocacy.

Future Research Aims

Designing Data Futures that Evolve Individual Workers. Outside labor relations, workers and consumers of various domains produce value and data for platforms. As AI permeates knowledge and creative work domains, future generations will require new tools and frameworks to augment their unique human capabilities (*e.g.*, intellect [d], creativity, empathy) while preserving humanistic values (*e.g.*, dignity, autonomy, meaning) – when collaborating and creating with AI. In support of creative professionals, I advised an HCI+AI team to design and evaluate [POET](#), which identifies and expands visual homogeneities of text-to-image models to **automatically generate more diverse outputs** [9], and is extensible to modalities beyond images. I am eager to continue exploring techniques that preserve user agency and intent in early-stage pipelines.

In relatively service-based, low-income and vulnerable forms of work, many latent stressors and needs remain unacknowledged. As labor increasingly modularizes, emergent sensing capabilities (*e.g.*, video, speech) offer more lightweight methods for capturing tacit knowledge and human expertise. Combining methods from labor studies and organizational management with techniques from data science (*e.g.*, embedded design, visualization), I aim to develop richer, more effortless mechanisms that *visibilize, legitimize and revalue labor*.

Reconfiguring *Social Relations* to Motivate Digital Collective Actions. Gamification [10] represented one method of engaging consumer solidarity with workers through immersion. Another way to promote unity is to collectively recognize that *gig workers are also consumers* who benefit from platform resources – e.g. client pools, matching algorithms. I plan to apply frameworks from critical computing and behavioral economics to analyze and design mechanisms/relations that mobilize more bottom-up, consumer-led accountability.

Beyond gig labor markets, labor protections are falling behind accelerating technological growth and evolving worker needs, causing union membership to decline (currently <10% of US adults [6]). I will take a multi-stakeholder perspective to explore novel interventions – leveraging creative and novel techniques such as game or mechanism design – to better align workers and their allies toward scalable peer support.

Building Collaborative, Pluralistic and Trust-based *Communities*. Online communities of today (e.g. social media, open source development) suffer from misaligned incentives and disruption – e.g. inappropriate comments [4]. Trusting, sustainable communities require shared understandings of on-the-ground realities, negotiations to reconcile pluralistic values. PolicyCraft facilitates deliberation using human and AI-generated cases [8]. Many emergent capabilities (e.g., agent-based simulations, world-building) carry potential to scalably transform governance and our understandings of each other. I intend to investigate more unremarkable [1] systems that promote more prosocial, deliberative and inclusive communities, starting with familiar domains (e.g. gig work, art [12]) before exploring newer realms – e.g. music, moderation, game design [10].